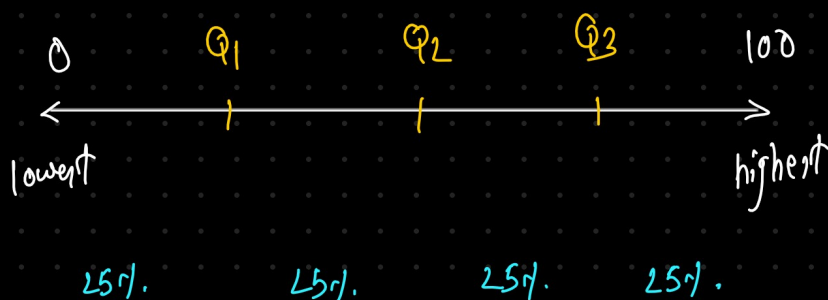
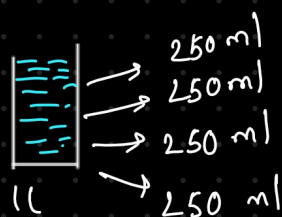


26-07-2025

Agenda - Statistics - III

Quartile → divide a data into 4 groups.

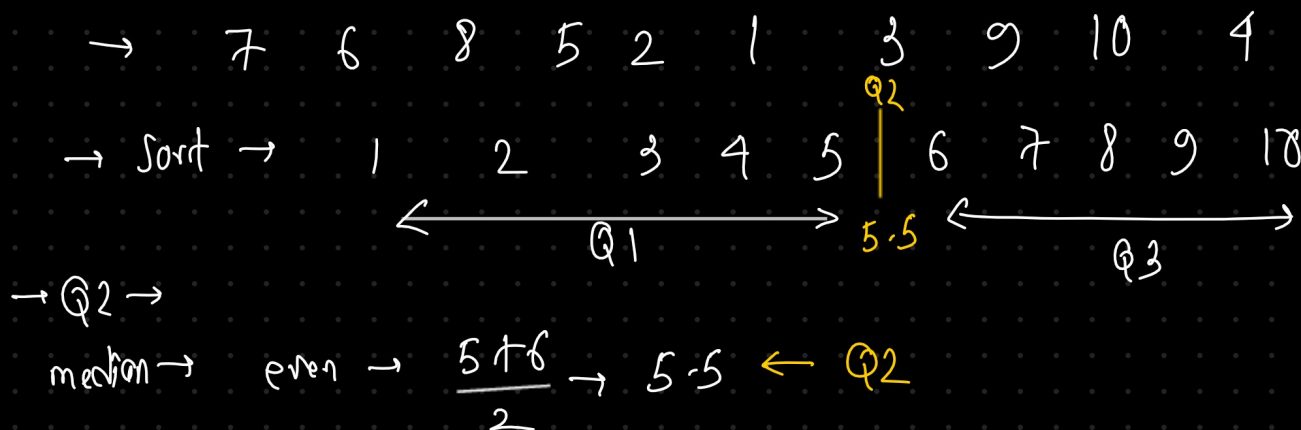


Percentage - A mathematical unit of measurement that displays the answer out of a total of 100.

Percentile - Percentile is a value from which the values of percentages below it are found. A percentile is based on the rank of numbers. The unit of percentile is denoted by xth, for example, 30th,

$Q_1 \rightarrow 25^{th}$ %tile
 $Q_2 \rightarrow 50^{th}$ %tile
 $Q_3 \rightarrow 75^{th}$ %tile

- Percentage is about your score.
- Percentile is about your rank among others.



median of values that fall below $Q_2 \Rightarrow Q_1 \rightarrow 3$
median of value that fall above $Q_2 \Rightarrow Q_3 \rightarrow 8$

IQR $\Rightarrow$ difference between 3rd & 1st quartile

$$\Rightarrow Q_3 - Q_1 \Rightarrow 8 - 3 \rightarrow \textcircled{5}$$

IQR is used to identify outliers

← 1 2 3 4 5 6 7 8 9 10 120 →

$$Q_2 \rightarrow 6$$

$$Q_1 \rightarrow 3$$

$$Q_3 \rightarrow 9$$

$$IQR \rightarrow 9 - 3 \Rightarrow 6$$

$$\text{IQR} \times 1.5 \rightarrow \textcircled{9}$$

$$Q_1 - (IQR \times 1.5) \rightarrow (3 - 9) \rightarrow \textcircled{-6}$$

$$Q_3 + (IQR \times 1.5) \rightarrow 9 + 9 \rightarrow \textcircled{18}$$

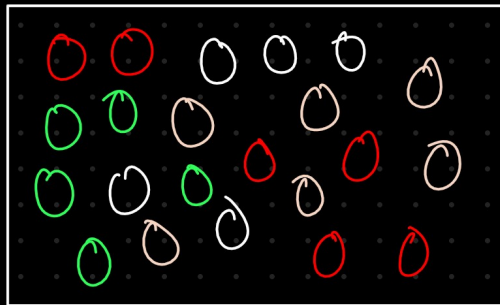
1 2 3 4 5 6 7 8 9 10 120

all under the range

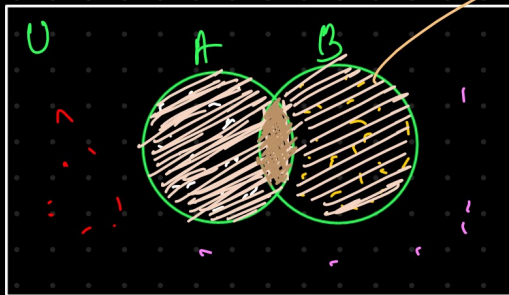
outlier

Set theory →

A set is a collection of distinct object, grouped as an entity



→ Set → U (universal set)
→ elements

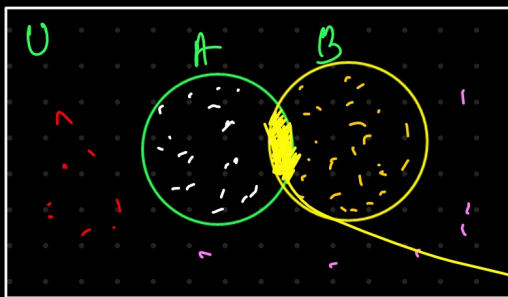


$A \cup B$

$A \cup B$

→ elements that belong to either A or B or both

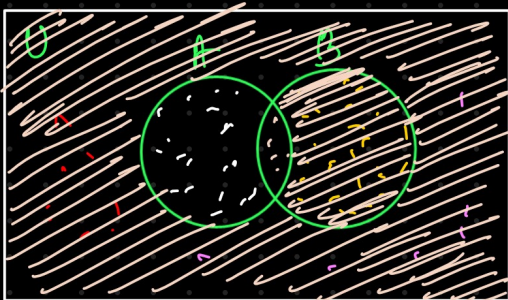
$\textcircled{A}$, $\textcircled{B}$,



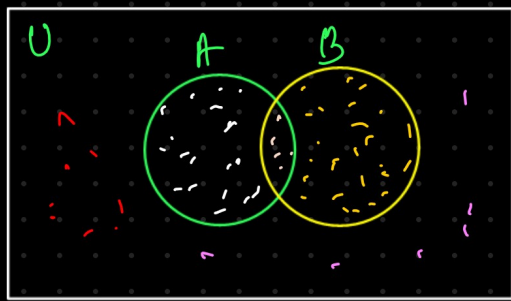
A intersect B

Both $\textcircled{A}$ & $\textcircled{B}$

$A \cap B$

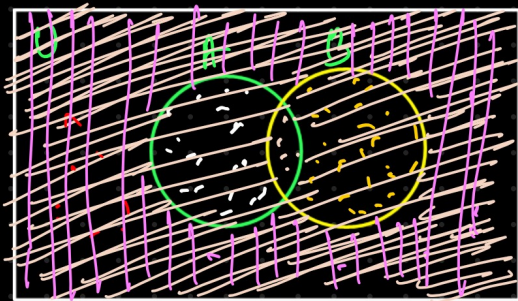
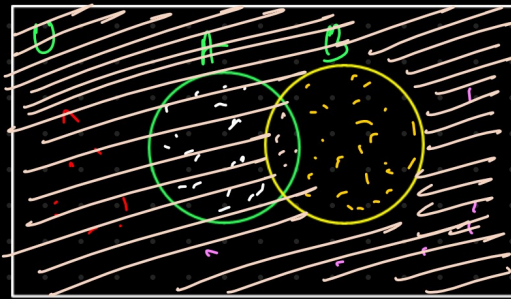
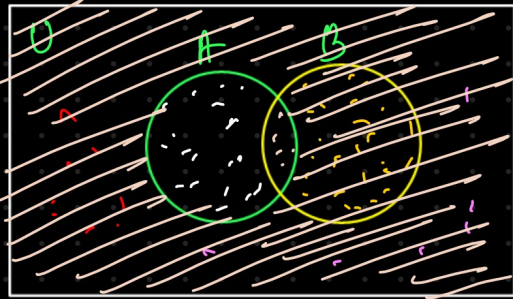


A^c → element that does not belong to A.



$$A' \cap B'$$

→



$$\rightarrow (A \cup B)'$$

$$A \rightarrow 1, 2, \boxed{3, 4}$$

$$B \rightarrow \boxed{3}, \boxed{4}, 5, 6$$

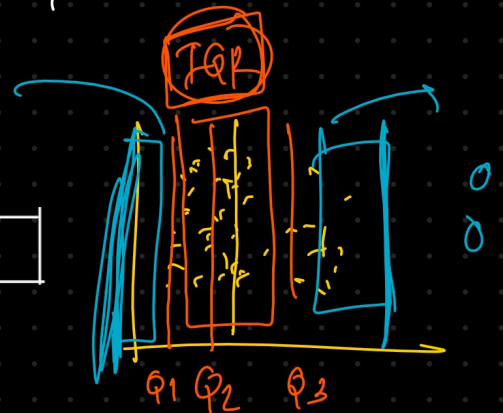
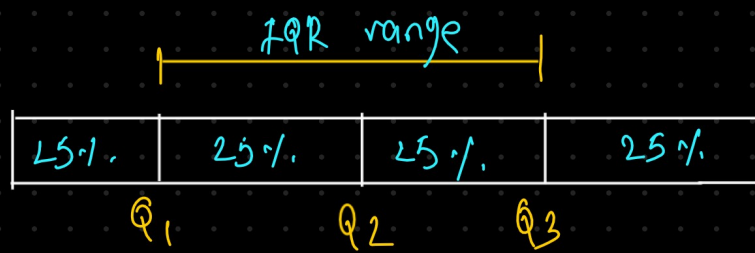
$$A - B = \boxed{A - (A \cap B)}$$

$$= \underline{\{1, 2, 3, 4\}} - \underline{\{3, 4\}} \rightarrow \{1, 2\}$$

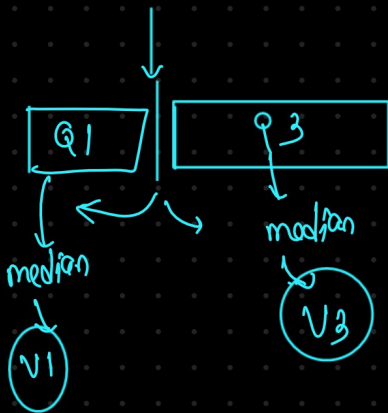
Recap:

(1) Range of values which will help us identify outlier.

IQR $\rightarrow$ value $\rightarrow$ 50% of the data



$Q_2 \rightarrow$ median $\rightarrow V_2$



$$IQR = Q_3 - Q_1$$

IQR

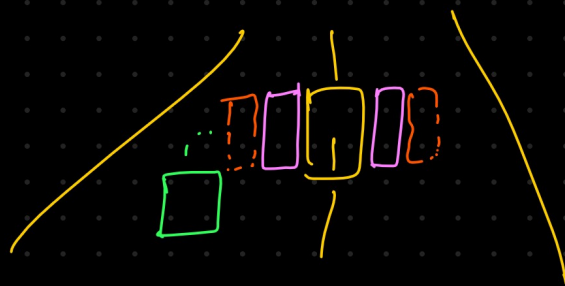
IQR

$$\text{range-in-left} = Q_1 - (1.5 * IQR) \rightarrow$$

$$\text{range-in-right} = Q_3 + (1.5 * IQR) \rightarrow$$

before

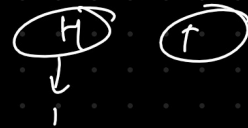
after



more sensitive range

Q. Tossing a coin.

Event: — Get Head $\rightarrow 1$



$$P(\text{Head}) = \frac{1}{2} = 0.5$$

Q. Rolling a dice

$\rightarrow \{1, 2, 3, 4, 5, 6\} \rightarrow 6$

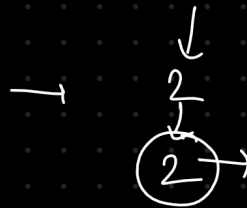
$\rightarrow$ Event, getting a 4 $\rightarrow 1$

$$P(4) = \frac{1}{6} = 0.16$$

Q. Rolling a faulty dice

$\{1, 2, 3, 2, 4, 5\}$

event $\rightarrow 2 \rightarrow 2$



$$P(2) = \frac{2}{6} = \frac{1}{3} = 0.33 \uparrow$$

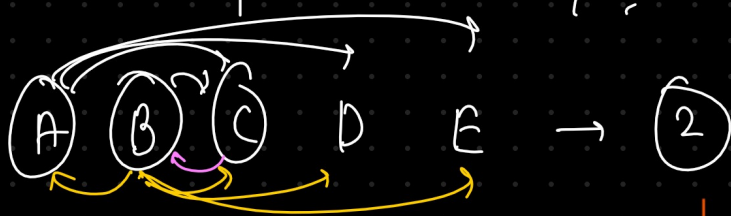
$\rightarrow$ Note:

$\rightarrow$ probability can never be negative or > 1

$\rightarrow$ sum of all outcomes = 1

Q. Choose committee member

* 2 students, 5 student, ? How many way i have to pick 2 student,



AB, AC, AD, AE

~~BA=AB~~, BC, BD, BE

~~CB=BC~~, ~~BA=AB~~, CD, CE

~~DC~~, ~~DB~~, ~~DA~~, DE

10

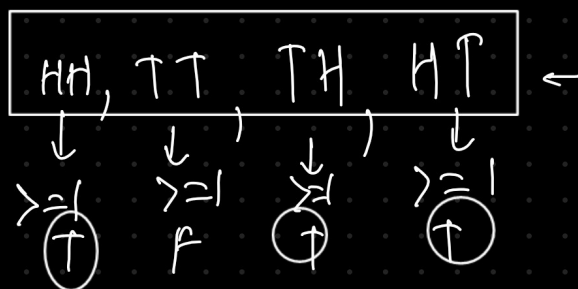
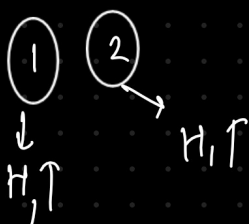
$n=5$
 $k=2$

AB & BA
repetition

$${}^5C_2 = \frac{5!}{2!(5-2)!} \rightarrow \frac{5 \times 4 \times 3 \times 2}{2 \times 3 \times 2} = 10$$

$\rightarrow \frac{3!}{3 \times 2} = 1$

Q. Tossing 2 coin



≥ 1

Event $\rightarrow$ H count == 1

$\rightarrow$ favorable outcomes $\rightarrow 2$

total $\rightarrow 4$

$$p(=1 \text{ Head}) = \frac{2}{4} = \frac{1}{2} = \underline{0.5}$$

Q. 2 coin

Event $\rightarrow$ (Atleast 1 H) $\rightarrow$

$f = 3$

$t = 4$

$$\frac{3}{4} = \underline{0.75} = p(\geq 1 \text{ Head})$$

Q. Tossing 3 coin

Event $\rightarrow$ Atleast 2 Heads

$$= 2 > 2$$

$\geq 2 \rightarrow$

	c1	c2	c3
<u>T</u>	H	H	H
F	T	T	T
<u>T</u>	H	T	H
F	T	H	T
<u>T</u>	H	H	T
F	T	T	H
<u>T</u>	H	T	T
F	T	H	H

$\rightarrow$ outcomes

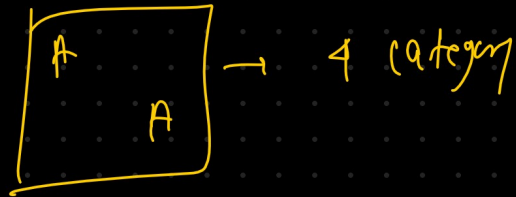
$$\rightarrow P = \frac{4}{8} = \frac{1}{2} = \underline{\underline{0.5}}$$

Total outcomes $= 2^n$
 $= 2^3 \rightarrow 8 \rightarrow 2 \times 2 \times 2$

Q. Draw 1 card from 52 cards

$P(\text{Ace})$

Ace $\rightarrow 4$



$$\frac{F}{T} = \frac{4}{52} = \frac{1}{13} = \underline{\underline{0.076}}$$

Q. Draw 2 cards

Ace

$$52 \rightarrow 1 \rightarrow P() \rightarrow 4/52$$

$$52 \rightarrow 1 \rightarrow P() \rightarrow 4/52$$

$$\left(\frac{4}{52}\right) \times \left(\frac{4}{52}\right) \times 0.005$$

Note:

$P(E1)$ And $P(E2) \rightarrow *$

$P(E1)$ or $P(E2) \rightarrow +$

p. $\rightarrow$ probability of getting an Ace and getting another Ace in second pick

$$\textcircled{N} \rightarrow \textcircled{52} \rightarrow 1 \rightarrow \frac{1}{52} \rightarrow \textcircled{1}$$

$$\rightarrow \textcircled{52} \rightarrow 1 \rightarrow \frac{1}{52} \rightarrow$$

$$\frac{1}{52} \times \frac{1}{52} \rightarrow$$

